
Symmetries in Physics — Exercise Sheet 12

Integration, characters, and Clebsch–Gordan

Variant: Questions only

Exercise 1 On $SU(2)$ the class of U is fixed by $\theta \in [0, 2\pi]$ with $\text{tr } U = 2 \cos(\theta/2)$, and the class integration formula (the *Weyl integration formula*) reduces class-function integrals to

$$\int_{SU(2)} f = \frac{1}{\pi} \int_0^{2\pi} f(\theta) \sin^2(\theta/2) d\theta.$$

Verify the normalisation, and prove by direct integration that the characters $\chi_j(\theta) = \sin((2j+1)\theta/2)/\sin(\theta/2)$ are orthonormal.

Exercise 2 Prove the Clebsch–Gordan series $j_1 \otimes j_2 = \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j$ by weight counting alone.

Exercise 3 Compute all Clebsch–Gordan coefficients of $1 \otimes \frac{1}{2} = \frac{3}{2} \oplus \frac{1}{2}$ for the weights $m = \pm \frac{1}{2}$, by ladders and orthogonality, in the Condon–Shortley convention. Here that convention means: keep the printed factor order; construct coupled multiplets in descending j ; make the largest- m_1 product coefficient of each new highest-weight state positive and real; then use the positive total- J_- ladder. Reversing one whole multiplet is equivalent unless the exact adopted phase is requested, as it is here.

Exercise 4 Prove the projection theorem: within a multiplet $|\alpha jm\rangle$ with $j > 0$, every vector operator $\vec{V}$ satisfies

$$\langle \alpha jm' | V_q | \alpha jm \rangle = \frac{\langle \vec{J} \cdot \vec{V} \rangle_{\alpha j}}{j(j+1)} \langle jm' | J_q | jm \rangle,$$

and derive the Landé factor $g_J = 1 + \frac{j(j+1) + s(s+1) - \ell(\ell+1)}{2j(j+1)}$.

Exercise 5 Derive the selection rules for the Gaunt-type integral $I = \int Y_{\ell_3 m_3} Y_{\ell_2 m_2} Y_{\ell_1 m_1} d\Omega$: it vanishes unless $m_1 + m_2 + m_3 = 0$, the triangle inequality $|\ell_1 - \ell_2| \leq \ell_3 \leq \ell_1 + \ell_2$ holds, and $\ell_1 + \ell_2 + \ell_3$ is even. (Use that $\overline{Y_{\ell m}}$ again lies in $\mathcal{Y}_\ell$, with L_z -weight $-m$, so that I is, up to a phase, the matrix element $\langle Y_{\ell_3, -m_3} | Y_{\ell_2 m_2} | Y_{\ell_1 m_1} \rangle$.)

Exercise 6 Construct the Haar integral of $SO(3)$ from that of $SU(2)$ through the double cover Φ , prove its invariance, and derive the class measure

$$\int_{SO(3)} f = \frac{2}{\pi} \int_0^\pi f(\theta) \sin^2(\theta/2) d\theta$$

for class functions of the rotation angle.